

$$\begin{aligned}
SKK &= ((\lambda xyz.xz(yz))K)K \\
&\rightarrow_{\beta} (\lambda yz.Kz(yz))K \\
&\rightarrow_{\beta} \lambda z.Kz(Kz) \\
&= \lambda z.(\lambda xy.x)z(Kz) \\
&\rightarrow_{\beta} \lambda z.z \\
&= I
\end{aligned}$$